IB PHYSICS E.3 RADIOACTIVE DECAY

Isotopes

- Same **proton number Z**, different **neutron number N**.
- **Nucleon number** A = Z + N.
- Same **chemical** properties, different **physical** stability.
- **Evidence for neutrons**: isotopes with same Z but different A.

Mass-energy equivalence

- In **nuclear reactions**, missing mass = energy released.

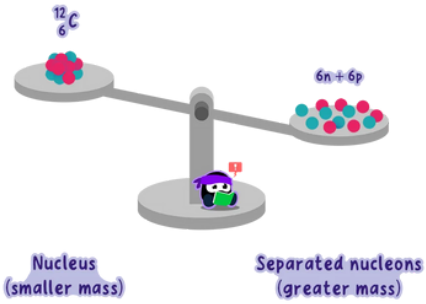
$$E = mc^2$$

- Applies to **fission, fusion, and decay**.

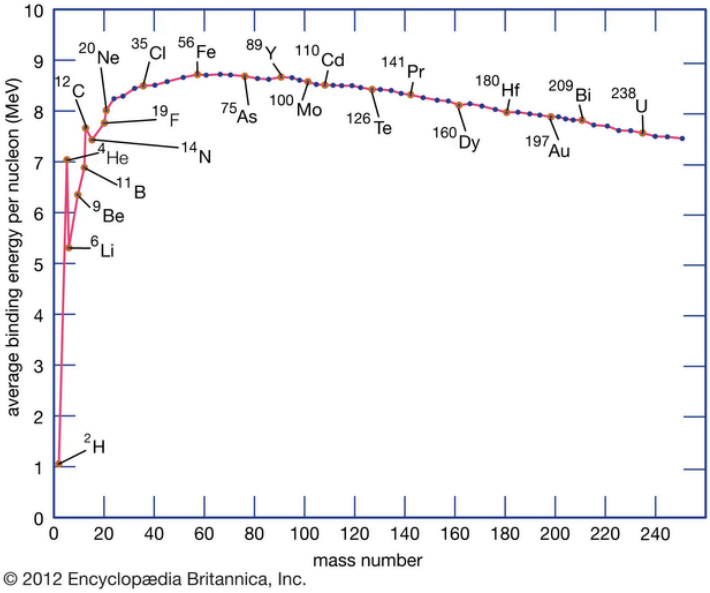
Nuclear binding energy & mass defect

- Mass of nucleus < sum of separate nucleons.
- **Mass defect Δm** → converted into binding energy:

$$E = \Delta mc^2$$



- **Binding energy per nucleon** → measure of stability.
- Increases rapidly for light nuclei → peak at Fe-56.
- Fusion (light nuclei → heavier) → releases energy.
- Fission (heavy nuclei → lighter) → also releases energy.
- Beyond A ≈ 60 → stability decreases.



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Nuclear binding energy. Source: https://cdn.britannica.com/46/6046-050-D533C3B3/energies-function-atomic-mass-number.jpg:

Strong nuclear force

- **Strong nuclear force** = short-range (~1 fm), **attractive** force between all nucleons.
- Overcomes **proton-proton** electrostatic repulsion inside the nucleus.
- **Independent of charge** — acts equally between p-p, n-n, p-n.
- Evidence from **high-energy nucleon scattering** experiments: attractive at short range, negligible beyond ~3 fm.



Note!

Without this force, nuclei would not exist.

Penetration & ionization (HL)

- **Alpha:** highly ionizing, low penetration (paper/air stops).
- **Beta:** moderate ionization, penetration (thin Al stops).
- **Gamma:** low ionization, high penetration (lead/concrete).



Remember!

Order of penetration: γ > β > α.

Types of radioactive decay

- **Alpha (α):** emits He nucleus → Z − 2, A − 4.
- **Beta minus (β^-):** neutron → proton + electron + antineutrino → Z + 1, A same.
- **Beta plus (β^+):** proton → neutron + positron + neutrino → Z − 1, A same.
- **Gamma (γ):** high-energy photon emitted → Z, A unchanged.

Neutrinos and energy conservation (HL)

- **Neutrinos (ν)** and **antineutrinos ($\bar{\nu}$)** proposed by Pauli to conserve **energy & momentum** in β decay.
- Neutral, tiny mass, weakly interacting → hard to detect.
- Presence explains **continuous β energy spectrum**.

Random & spontaneous nature of decay (HL)

- **Random:** cannot predict which nucleus decays.
- **Spontaneous:** unaffected by external factors (T, P, chemical bonding).

Radioactive decay law, decay constant and half-time (HL)

- **Decay constant λ** = probability of decay per nucleus per unit time.
- Number of **undecayed nuclei** over time:

$$N(t) = N_0 e^{-\lambda t}$$

- **Activity** (A)

$$A = \lambda N = \lambda N_0 e^{-\lambda t}$$

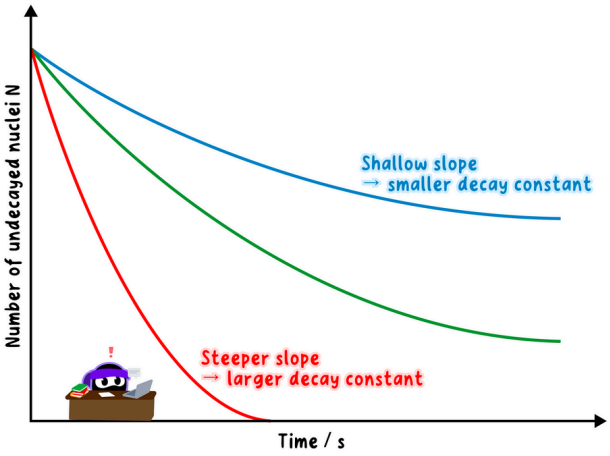
- **Half-life T½** = time for half the nuclei to decay:

$$T_{1/2} = \frac{\ln 2}{\lambda}$$



Exam Tip!

Larger λ → shorter $T_{1/2}$



Stability, N/Z ratio and binding energy curve (HL)

- **Stability** depends on **neutron-proton ratio**:
  - Light nuclei stable with N ≈ Z.
  - Heavy nuclei require N > Z.
- Too many neutrons → β^- decay; too few → β^+ decay.
- Above A ≈ 60, **binding energy per nucleon remains roughly constant** → fusion of heavy nuclei doesn't increase stability.

Discrete and continuous nuclear spectra (HL)

- **Alpha & gamma spectra**: **discrete energy levels** → evidence nuclei have quantized energy states (like atoms).
- **Beta spectrum**: **continuous**, due to energy sharing between electron/positron and neutrino.



Note!

- **Gamma** → no change to A or Z, just energy level change.
- **Alpha particles** from a given nuclide have characteristic energies.